

Rajeev K R

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Professional Summary

IIT Hyderabad alumnus and R&D Mechanical Engineer currently pursuing an M.Tech. in Computational and Data Sciences at IISc, with a strong foundation in mathematics, numerical computation, machine learning, and data science. Experienced in applying computational and mathematical methods to engineering and machine learning problems, with hands-on proficiency in PyTorch, TensorFlow, and scikit-learn. Interested in research at the intersection of numerical methods, optimization, and the mathematical foundations of machine learning.

Scholastic Achievements

- **GATE DA (2026):** General Rank - 263, GATE Score - 727
- **JEE (2020):** Advanced General Rank - 7190, Main General Rank/Percentile - 16368/98.53%

Technical Skills

- **Programming & ML Frameworks:** Python (Expert), PyTorch (Advanced), TensorFlow (Intermediate), Scikit-learn (Advanced), C/C++ (Intermediate), MATLAB (Intermediate)
- **Engineering Tools:** SolidWorks (Expert), Ansys Mechanical (Advanced), LaTeX (Beginner)

Patents/Awards

- **Industrial Design Patent:** Crew Member Unit – A compact network user interface system, CGPDTM. Filed in Mar 2026 (Grant Awaited).
- **Utility Patent:** Novel Design of a Pair of Add-on Mating Attachments with User Configurable Key Coding for Electrical Connectors, CGPDTM (Yet to be reviewed and filed by attorney).

Experience

Bharat Electronics Limited

Deputy Engineer (PD&IC Division)

Bangalore, India

Aug 2024 – Jul 2026

- Leading the complete mechanical lifecycle of indigenous products from concept design to structural/thermal simulations.
- Coordinating vendor procurement, prototyping, in-house and customer-site testing, validation, and long-term product support.
- Voluntarily took up an ML project with the objective of replacing engineering simulation software in the future. As an initial step, an ML model for predicting frequency of first mode of vibration of PCBs was developed and deployed as a web-app in company's local network for any engineer to use.

Education

Indian Institute of Science, Bangalore

M.Tech in Computational and Data Sciences

Bangalore

2026 – present

Indian Institute of Technology Hyderabad

B.Tech in Mechanical Engineering

CGPA: 8.09 / 10

Hyderabad

2020 – 2024

Govt. Model Boys Higher Secondary School

Higher Secondary (Physics, Mathematics, Chemistry, Biology, English, Hindi)

Percentage: 98.08%

Kunnamkulam, Kerala

2017 – 2019

M V M Residential Higher Secondary School

SSLC

Grade: Full A+

Valayamkulam, Kerala

2016 – 2017

Independent Research & Scholastic Studies

Mathematical Formulation of Adaptive Step-Size Transformations to improve SOTA optimizers

Jul 2026 – Present

- Investigating continuous, non-linear step-size transformations to control gradient scaling and momentum stability in high-dimensional optimization landscapes.
- Standard adaptive optimizers, such as AdamW, update parameters by tracking the first and second raw moments, m_t and v_t , using exponential moving averages. The Lion optimizer simplifies the first moment further by just keeping the sign, not the magnitude. The former excels in taking precise steps, whereas the latter excels in lesser memory and compute overhead at the cost of not being as precise with each step as the former. But this advantage lets the machine compute higher number of iterations for the same time/cost, helping the model to achieve similar or better results with Lion if tuned properly.
- The objective of my independent research is to invent a new algorithm that sits on a different point of this precision vs computational cost trade-off curve that can potentially outperform these SOTA optimizers.

Publications

A Machine Learning based approach in finding the Frequency of electronic PCB

Published in the internal technical magazine at PDIC, Bharat Electronics Limited; Yet to be published in an external conference

Projects

- **A Machine Learning based approach in finding the Frequency of electronic PCB** (Dec 2025): Analyzed and benchmarked 12 ML models (including neural networks and transformer-based pretrained models) to predict the frequency of the first mode of vibration of PCBs based on length, width, and copper percentage. Selected ExtraTreesRegressor as the most suitable model. Planning to shift the research to Gaussian Process Regression and other bayesian models.
- **911 Calls Analysis in the US** (May 2023): Comprehensive analysis using K-means clustering techniques employing heatmaps, scatterplots, and bar diagrams. ([View on GitHub](#))
- **Multi-utility crawler robot using Arduino** (Mar 2023): Designed a 6-limbed spider-bot using plastic 3D printing and Arduino for control and mobility.
- **Chess game** (Jan 2024): Custom chess game built in Python using the Pygame library. ([View on GitHub](#))
- **Plate Bounce** (Dec 2023): Custom game in Python using the Pygame library. ([View on GitHub](#))
- **Maze Generation Algorithm** (Jun 2023): Implemented a random maze generation algorithm using stack and recursive backtracking in C++. ([View on GitHub](#))

Relevant Coursework

Ongoing coursework

- Numerical Linear Algebra, Stochastic Models and Applications, Numerical Methods, Introduction to Scalable Systems, Introduction to Data Science

Completed coursework

- **Mathematics:** Calculus, Elementary Linear Algebra, Differential Equations.
- **Computer Science & AI:** Introduction to Artificial Intelligence, Introduction to Programming, Data Science Analysis, Image and Video Processing, Modelling and Simulation.

Volunteer Experience

National Service Scheme (NSS)

IIT Hyderabad | 2020 – 2024

Extracurricular Interests

- Self-taught beginner-level piano, guitar, and violin player.
- Football and fitness enthusiast.